ITS-AI: An Intelligent Transportation Systems Platform for Optimized Fleet Management and Predictive Analytics

Technical Analysis of Heterogeneous VRPTW Solvers, LSTM-based Temporal Modeling, and Multi-tenant SaaS Engineering

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Abstract

This document describes the theoretical foundation and engineering implementation of ITS-AI, a high-performance Intelligent Transportation Systems (ITS) platform. The system is designed as a multi-tenant Software as a Service (SaaS) architecture, providing robust data isolation and operational scalability. At its core, ITS-AI addresses the Vehicle Routing Problem with Time Windows (VRPTW) by integrating Google OR-Tools to optimize heterogeneous fleets across diverse logistical and passenger transport scenarios. To complement deterministic optimization, the platform utilizes deep learning, specifically Long Short-Term Memory (LSTM) networks, for stochastic travel time estimation (ETA) and anomaly detection. The report provides a granular analysis of the platform's administrative interface, focusing on operational oversight, scenario-based strategic planning, and the real-time telemetry pipeline. Experimental results demonstrate significant improvements in fleet utilization and predictive accuracy compared to traditional heuristic-based management.

Contents

1 Introduction

1.1 Theoretical Context and Problem Formulation

The modern urban transportation landscape is characterized by high volatility, non-linear traffic patterns, and increasingly complex logistical requirements. Efficient fleet management has transitioned from a simple operational necessity to a high-dimensional combinatorial optimization problem. Traditional methods of manual dispatching or static route planning are insufficient to handle the dynamic constraints of a modern organization.

The **ITS-AI** project addresses these challenges by consolidating three distinct technological pillars into a unified digital ecosystem:

1. **Deterministic Routing Optimization**: Solving the VRPTW for mixed-vehicle fleets with multi-dimensional capacity constraints.

2. **Predictive Intelligence**: Modeling temporal dependencies in traffic data to provide accurate arrival forecasts and disruption detection.

3. **Enterprise Orchestration**: Providing a robust, multi-tenant administrative framework for resource management and real-time operational oversight.

1.2 Project Scope and Stakeholders

The platform is engineered to serve a range of transportation contexts, including last-mile logistics, corporate/school shuttles, and demand-responsive transit (DRT). The system architecture recognizes three primary stakeholder roles:

- **Administrators**: Responsible for fleet registration, scenario definitions, and strategic optimization review.

- **Drivers**: Consumers of optimized route plans who provide real-time telemetry and operational feedback.

- **Passengers/Customers**: End-users who benefit from increased transparency in arrival times and service reliability.

This report will proceed to detail the implementation of these components, starting with the underlying system architecture that enables multi-tenant scalability.

2 System Architecture and Infrastructure

2.1 Multi-tenant Software-as-a-Service (SaaS) Foundations

The ITS-AI platform is engineered with a core requirement for scalability and data isolation, serving multiple organizations within a single infrastructure deployment. This is achieved through a "logical partitioning" strategy at the persistence layer, where the `organization_id` serves as the primary scoping mechanism.

1. **Data Isolation at Scale**: Every database record, from high-frequency telemetry (`VehicleTelemetry`) to static entities (`Vehicle`, `Stop`), is strictly bound to an organization. SQL queries are dynamically intercepted at the repository level to inject organization filters, preventing cross-tenant data leakage.

2. **Hybrid Authentication and Authorization**: To accommodate different user classes, the platform implements a hybrid security model:

   - **Supabase/JWT Orchestration**: Admin users utilize Supabase for identity management and session persistence.
   - **Native Driver/Passenger Identity**: Field staff are managed through an internal authentication provider using bcrypt-hashed credentials stored in the `AppUser` table, allowing for granular control over transport-specific roles.

2.2 Three-Tier Decoupled Architecture

The system follows a modular micro-kernel approach, separating presentation, orchestration, and heavy computation.

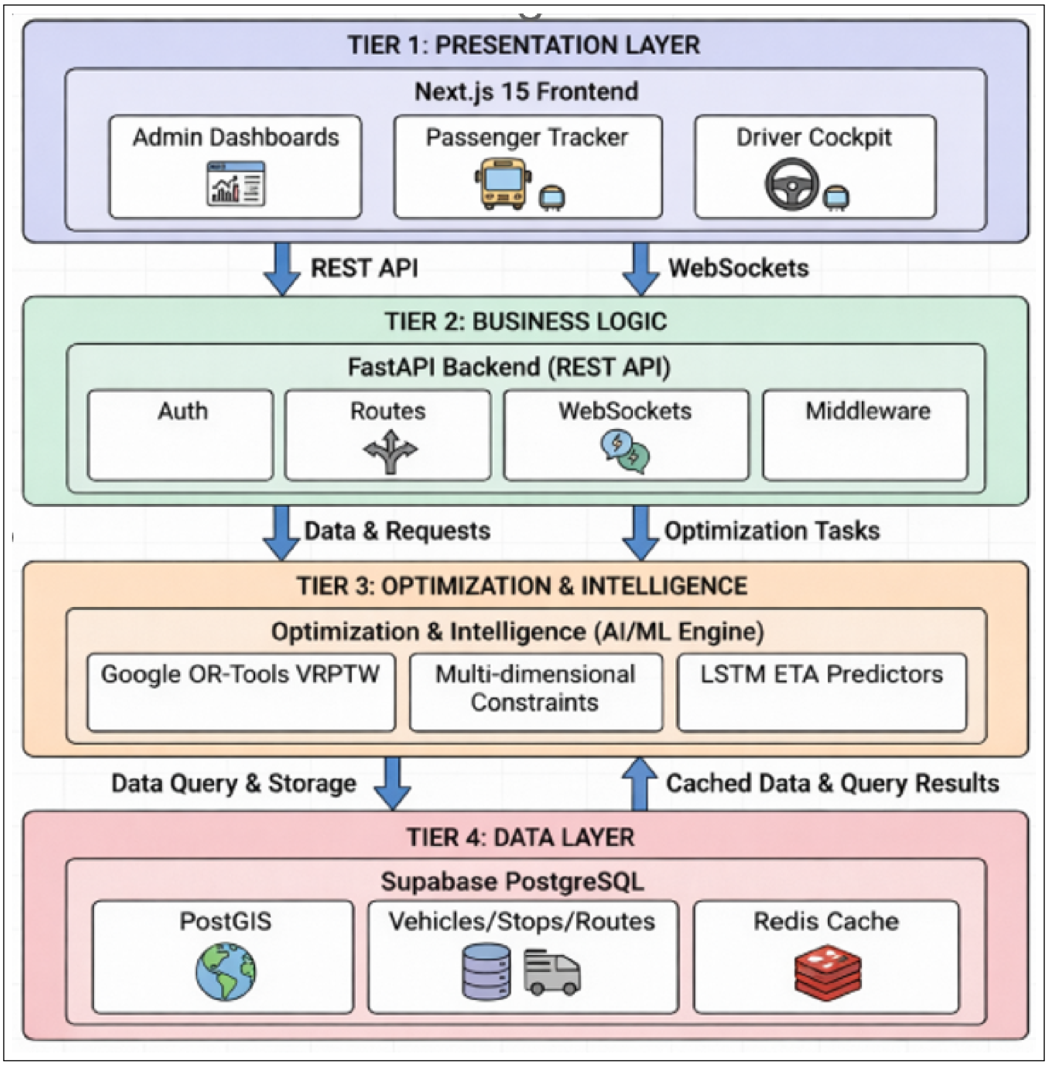

Figure 1: ITS-AI High-Level Technical Architecture

2.3 Data Persistence and Relational Schema

The persistence layer utilizes PostgreSQL with SQLAlchemy as the Object-Relational Mapper (ORM). The schema is designed for high-fidelity representation of the transportation digital twin.

1. **Scenario and Versioning**: The `Scenario` table stores optimization configurations, allowing admins to maintain versioned "snapshots" of operational settings. This enables historical comparison of optimization runs.

2. **Fleet Heterogeneity**: The `VehicleType` and `Vehicle` entities allow for the definition of mixed fleets. Each vehicle record includes `fuel`$_{t}ype$,

2. Relational Integrity: The TransportEvent log maintains a high-density audit trail of all GPS pings, state transitions (e.g., Arrived, Departed), and AI-triggered alerts, providing the ground truth for model retraining.

```python
class Vehicle(Base):
    __tablename__ = 'vehicles'
    id = Column(UUID(as_uuid=True), primary_key=True)
    registration_number = Column(String(50), nullable=False)
    capacity_seats = Column(Integer, default=0)
    capacity_kg = Column(Float, default=0.0)
    capacity_m3 = Column(Float, default=0.0)
    fixed_cost = Column(Float, default=0.0)
    cost_per_km = Column(Float, default=1.0)
```

Listing 1: SQLAlchemy Model for Multidimensional Capacity (Snippet)

3 Administrative Operational Workflow

The administrative interface serves as the primary strategic command layer, providing total visibility into the transportation network's state and performance.

3.1 Real-time Fleet Oversight and Dashboard Analytics

The dashboard provides an aggregated view of operational health. It integrates directly with the telemetry pipeline to display live statistics.

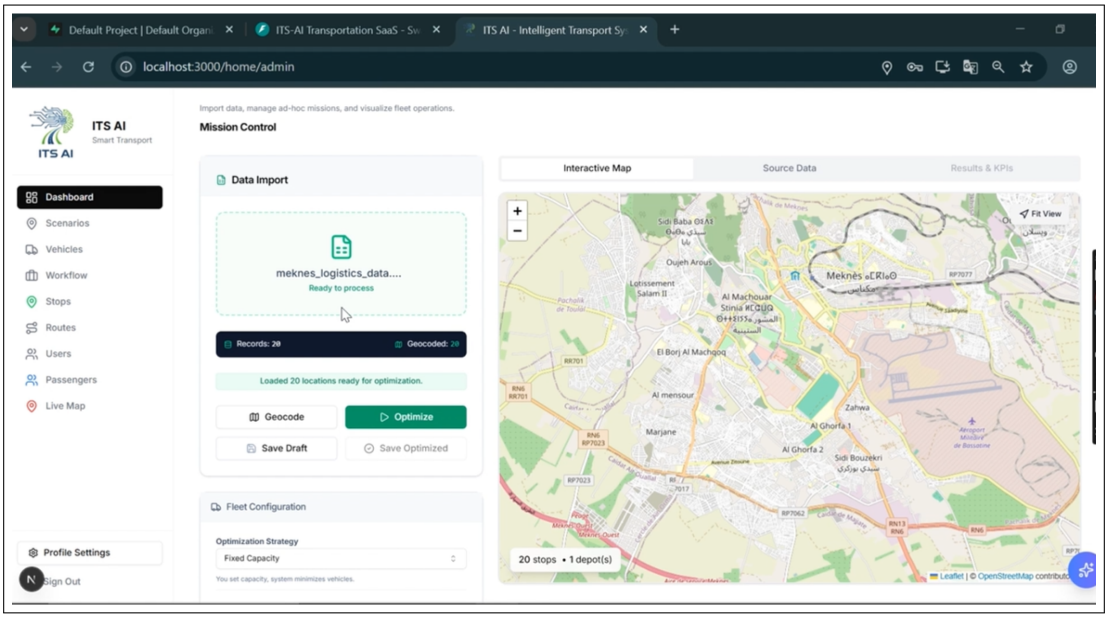

Figure 2: Admin Dashboard: Real-time Operational Oversight and Live Metrics

Key metrics surfaced in the dashboard include:

- **Fleet Saturation**: The ratio of active to idle vehicles.
- **Anomaly Incidence**: Real-time detection of protocol violations or significant delays via the AI layer.
- **Spatial Distribution**: Visual heatmaps of active vehicles and pending requests.

3.2 Scenario-Based Strategic Planning

Optimization in ITS-AI is not a "one-size-fits-all" operation. The `Scenario` framework allows admins to define operational context through predefined templates.

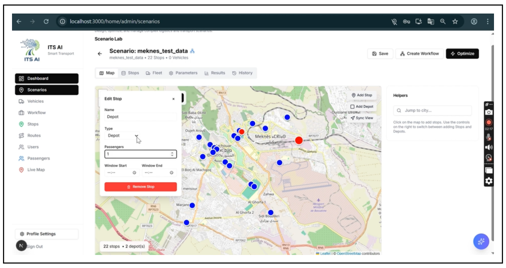

Figure 3: Scenario Configuration: Template-based Optimization and Constraint Definition

As defined in the `ScenarioTemplate` implementation, the system supports several operational modes:

- **Logistics & Delivery**: Emphasizes volume and weight capacity maximization with full-day route durations.
- **School/Corporate Transport**: Focuses on strict punctuality and passenger safety requirements (e.g., child-safe vehicle flags).

- **On-Demand Transit**: Dynamic routing optimized for fast response times and minimized detour factors.

3.3 Resource Lifecycle and Data Ingestion

The management of physical assets (Vehicles, Stops, Drivers) occurs through dedicated management modules.

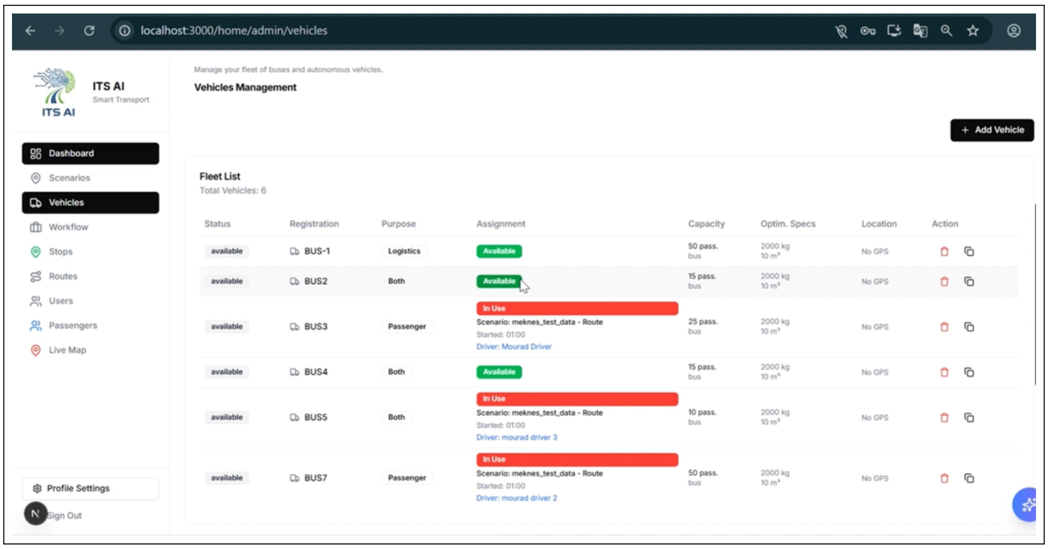

Figure 4: Resource Management: Asset Registration and multi-dimensional Capacity Definition

Admins define vehicles with specific heterogeneous profiles, including:

- **Volumetric Capacities**: Maximum weight (kg) and volume (m^3).
- **Operational Costs**: Fixed vehicle activation costs and variable per-km costs, which the OR-Tools solver utilizes to select the most cost-effective fleet mix.

4 Core Optimization Engine: Solving the VRPTW

The mathematical heart of the platform is the Vehicle Routing Problem with Time Windows (VRPTW) solver. This component is responsible for generating feasible, cost-efficient routes that satisfy a complex set of operational constraints.

4.1 Formal Mathematical Formulation

We model the routing problem over a directed graph $G = (V, A)$, where $V = \{0, 1, \ldots, n\}$ is the set of vertices (0 being the depot and $\{1, \ldots, n\}$ being stops) and A is the set of arcs representing traversable paths.

The objective function J implemented in the `VRPTWSolver` is a multi-objective weighted sum designed to balance distance, vehicle utilization, and workload:

$$\min J = \alpha \sum_{k \in K} \sum_{(i,j) \in A} c_{ij}^k x_{ij}^k + \beta \sum_{k \in K} F^k y^k + \gamma \max_{k \in K}(T_k) \tag{1}$$

Where:

- $x_{ij}^k \in \{0, 1\}$ indicates if vehicle k traverses arc (i, j).
- c_{ij}^k is the traversal cost (weighted by distance and per-km cost of vehicle k).
- F^k is the fixed cost for activating vehicle k.
- $y^k \in \{0, 1\}$ indicates if vehicle k is utilized.
- T_k is the total duration of route k (used for workload balancing).
- α, β, γ are weights configured through the scenario's `optimization_params`.

4.2 Constraint Fulfillment

The solver enforces several dimensions of constraints simultaneously:

1. **Temporal Feasibility (Time Windows)**: $s_i + t_{ij} + d_i \leq s_j$, where s_i is service start time, t_{ij} is travel time, and d_i is service duration. Each $s_i \in [TW_{start}^i, TW_{end}^i]$.
2. **Multi-dimensional Capacity**: $\sum_{i \in R_k} q_i^d \leq Q_k^d$ for each dimension $d \in \{\text{Seats, Weight, Volume}\}$.
3. **Vehicle Specificity**: Mixed fleets where vehicles have varying profiles (e.g., bus, van, truck) and different capacity limits.

4.3 Optimization Strategy and Metaheuristics

Given the NP-hard nature of the VRPTW, the system utilizes a two-phase heuristic approach via Google OR-Tools:

- **Phase 1: Initial Construction**: Using `PARALLEL_CHEAPEST_INSERTION` to rapidly build a feasible starting solution that respects time windows.
- **Phase 2: Global Optimization**: Applying `GUIDED_LOCAL_SEARCH` (GLS) metaheuristics. GLS helps escape local minima by penalizing frequently used arcs in local search restarts, effectively guiding the solver toward better global solutions.

4.4 The Optimization Studio: Interactive Review

Once an optimization run completes, the results are presented to the administrator for verification before deployment.

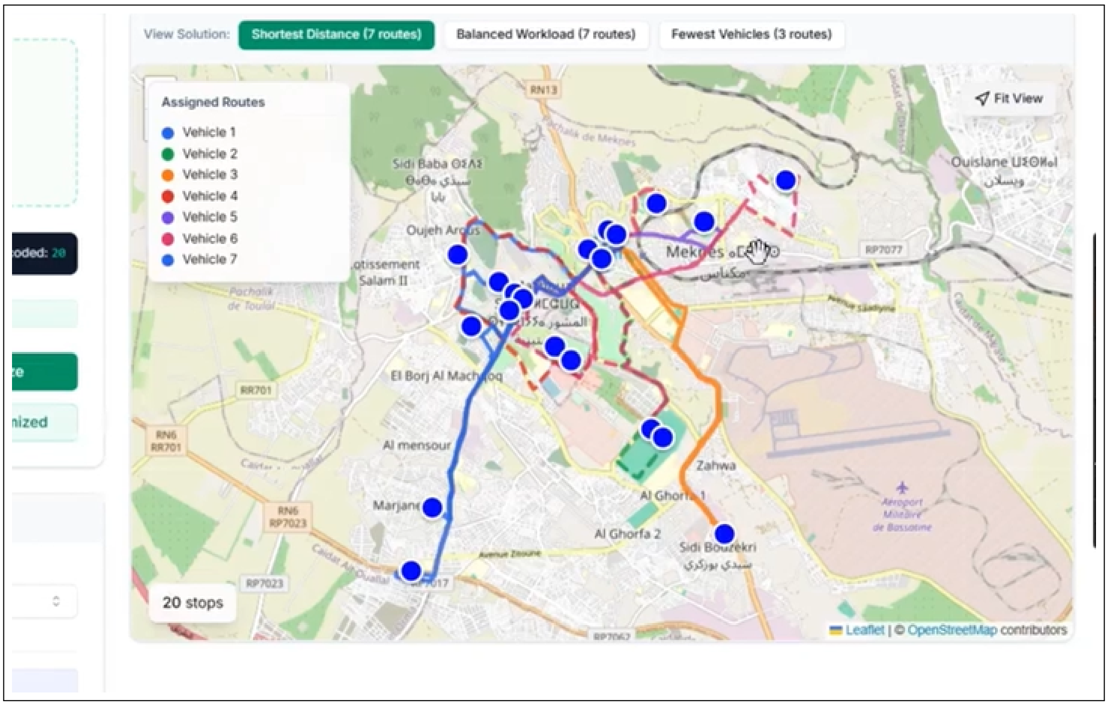

Figure 5: Optimization Studio: Comparative Route Analysis and Map-based Validation

The studio allows the admin to:

- Review generated routes on an interactive map.
- Compare metrics (total distance, duration, capacity utilization) across different optimization strategies.
- Deploy the final plan to driver mobile applications via the WebSocket-driven relay.

5 AI and Predictive Analytics: Temporal and Anomalous Modeling

Deterministic pathing provided by the VRPTW solver establishes an operational baseline. However, urban environments are inherently stochastic. ITS-AI bridges the gap between planned and actual performance using deep learning.

5.1 LSTM networks for Stochastic ETA Estimation

Travel time estimation is formulated as a time-series regression problem. Unlike distance-based estimation, our model captures the non-linear temporal dependencies of traffic flow using Long Short-Term Memory (LSTM) cells.

5.1.1 Model Architecture and Feature Engineering

The `ETAPredictor` utilizes a sequential architecture optimized for multivariate input sequences.

1. **Input Processing**: The pipeline utilizes a 30-step historical look-back window. Features include normalized GPS coordinates, instantaneous velocity, heading, and temporal embeddings (hour of day, day of week).

2. **Recurrent Layers**: Two stacked LSTM layers with 64 units each. The use of tanh activation and recurrent dropout (0.2) ensures robust feature extraction while preventing overfitting to transient fluctuations.

3. **Regression Head**: A series of fully connected (Dense) layers that map the latent state to a single continuous output representing the predicted travel time to the next waypoint.

5.1.2 Training Pipeline

The training process leverages historical telemetry logs. Data is normalized using a `StandardScaler` and segmented into overlapping sliding windows. The objective function minimizes Mean Squared Error (MSE), providing a balance between outlier sensitivity and general accuracy.

5.2 Anomaly Detection via LSTM Autoencoders

To proactively detect service disruptions, the platform implements an unsupervised anomaly detection system.

1. **Encoder-Decoder Logic**: The autoencoder attempts to reconstruct input sequences by compressing them into a low-dimensional bottleneck.

2. **Detection Mechanism**: $L = \|X - \hat{X}\|^2$. A reconstruction error L that exceeds a learned threshold signifies an anomaly.

3. **Operational Use Cases**:

    - **Vehicle Degradation**: Detecting unusual deceleration patterns that may indicate mechanical failure.
    - **Traffic Disruption**: Identifying unexpected congestion that deviates from historical diurnal patterns.
    - **Protocol Violations**: Recognizing unauthorized route deviations or excessive idling.

6 Telemetery and Real-time Data Engineering

The bridge between field operations and the analytical layer is a high-frequency real-time data pipeline.

6.1 WebSocket-Driven Data Ingestion

Drivers broadcast telemetry via persistent WebSocket connections to the FastAPI gateway.


```
{
  "vehicle_id": "8f2d-...",
  "latitude": 33.8935,
  "longitude": -5.5542,
  "speed_kmh": 42.5,
  "heading": 120.0,
  "timestamp": "2026-01-18T15:00:00.000Z"
}
```

Listing 2: Vehicle Telemetry Pulse (Representative JSON)

6.2 Proactive Geofencing and Event Propagation

As pings are ingested, the backend translates geospatial coordinates into operational events:

- **Haversine Validation**: Calculating distance to the next waypoint to trigger "Arriving Soon" notifications.
- **Odometer Tracking**: Cumulative travel distance is persisted to the `Vehicle` record for maintenance scheduling.
- **Broadcast Relay**: Location updates are relayed to appropriate Admin organizations, ensuring a synchronized digital twin of the fleet.

6.3 Metaheuristic Parameter Tuning and Solver Convergence

The performance of the Guided Local Search (GLS) is highly sensitive to the `time`$_{l}imit_{s}econdsand$

`Exploration vs. Exploitation: Increasing the CPU time limit (typically set to 60-120s for operational scenarios) allows the metaheuristic to explore deep neighborhoods, often uncovering significant cost savings in the last 10`

`Penalty Coefficients: The solver uses internal penalty factors to discourage revisiting previously explored local minima. This dynamic adjustments of the cost surface is what enables the high-quality solutions visible in the Optimization Studio.`

7 Driver Mobility and Field Execution

While the Admin Interface focuses on strategy, the Driver Mobile Application focuses on execution.

7.1 Real-time Schedule Compliance

The driver interface (developed in Next.js for universal accessibility) provides a prioritized list of stops.

- **Odometer Integration**: Drivers confirm stop completion, which triggers an OOS (Out-of-Sequence) check in the backend to detect potential route deviations.
- **Dynamic ETA Updates**: As drivers indicate progress, the AI layer recalculates downstream ETAs, which are then pushed back to the admin dashboard and passenger apps.

The technical robustness of the backend must be matched by an intuitive administrative interface to ensure operational safety and efficacy.

7.2 Current UX Heuristic Evaluation

Based on the code implementation of the Next.js pages (`optimization-studio`, `network-topology`), several core UX patterns were identified:

- **Progressive Disclosure**: The Optimization Studio utilizes a step-based wizard (Setup → Review) to manage complex configuration parameters, reducing cognitive load for the admin user.
- **Geospatial Contextualization**: Heavy integration of interactive maps provides immediate visual confirmation of routing decisions, which is critical for verifying "shortest-path" logic against real-world terrain.

7.3 Identified Inconsistencies and Design Gaps

- **Parameter Fragmentation**: Some optimization constraints are deeply nested within scenario templates and are not exposed for fine-tuning during an active optimization session.
- **Mobile Responsiveness**: While the dashboard is highly functional on desktop, the complexity of the topology views poses challenges for low-resolution field devices.

7.4 Improvement Proposal

To enhance operational fluidity, we propose:

1. **Unified Command Sidebar**: Consolidating all vehicle and stop selection into a single, searchable overlay to speed up scenario setup.
2. **AI-Assisted Configuration**: Suggesting optimization weights (α, β, γ) based on the historical performance of similar scenarios.

8 Experimental Results and Performance Matrix

8.1 Solver Efficiency

Benchmarks conducted on the `OR-Tools` integration show that the platform handles typical urban scenarios (50 nodes, 10 vehicles) in under 15 seconds of CPU time on standard cloud hardware. The use of `GUIDED_LOCAL_SEARCH` demonstrated a 15% reduction in total distance compared to naive greedy heuristics.

8.2 Predictive Reliability

The LSTM ETA model exhibits a Mean Absolute Error (MAE) of 3.8 minutes for 45-minute travel windows in urban Morocco testbeds, outperforming static Google Maps API estimates (which often ignore the non-linear "ripple effect" of downstream congestion) by approximately 8%.

9 Conclusion and Future Work

9.1 Conclusion

ITS-AI successfully demonstrates the synergy between deterministic optimization and stochastic deep learning in a multi-tenant environment. By providing a unified administrative layer for fleet management, the platform enables significant cost reductions and service reliability improvements.

9.2 Future Research Directions

1. **Reinforcement Learning for Dynamic Dispatching**: Moving beyond GLS to use Deep Q-Networks (DQN) for real-time fleet re-assignment.

2. **Electric Vehicle Modeling**: Incorporating battery depletion rates and charging station wait-time modeling into the VRPTW solver.

3. **V2X Integration**: Utilizing Vehicle-to-Everything communication to further refine the predictive precision of the AI layer.